

Geodesic Multidimensional Scaling: Self-Consistent Dimensionality Reduction via Graph Geodesic Distances

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Abstract

Multidimensional scaling (MDS) seeks a low-dimensional embedding of a set of points that faithfully reproduces a given pairwise dissimilarity matrix. Classical and metric MDS measure faithfulness via the discrepancy between the input dissimilarities and the *Euclidean* distances among the embedded points. Isomap improved upon MDS for curved-manifold data by replacing Euclidean input dissimilarities with shortest-path distances on a k -nearest-neighbor (k -NN) graph, thereby approximating intrinsic manifold geodesics. However, Isomap retains Euclidean distances on the *output* side, creating a fundamental asymmetry: input distances are geodesic, but the fidelity criterion is Euclidean. We argue that this asymmetry is not merely an aesthetic inconsistency but a systematic source of distortion whenever the data carry genuine graph or manifold structure.

We propose *Geodesic MDS*, a self-consistent formulation in which the input dissimilarities are graph geodesics and the output fidelity criterion measures the embedded lengths of those same input-side geodesics. The embedding is sought in $\mathbb{R}^d$ such that a fixed family of shortest paths chosen from the input k -NN graph retains its path-length geometry as closely as possible after embedding. We derive a closed-form gradient for this geodesic stress function, extend the SMACOF majorization framework to the geodesic setting, and characterize the role of k as a geometric scale parameter that continuously interpolates between a sparse tree-like regime and standard metric MDS. The resulting method is practically implemented by replacing one line in a standard MDS pipeline and adds no asymptotic complexity. We situate the proposal within the metric geometry literature—connecting it to minimum metric realizations, tight spans, and metric spanners—and discuss applications to high-dimensional compositional data, phylogenetic embeddings, and network visualization.

Keywords: multidimensional scaling, dimensionality reduction, geodesic distance, graph metric, k -nearest neighbor graph, Isomap, SMACOF, metric geometry, manifold learning, stress function

1 Introduction

The central task of *multidimensional scaling* (MDS) is to find a configuration of n points $\mathbf{Z} = (\mathbf{z}_1, \dots, \mathbf{z}_n) \in \mathbb{R}^{n \times d}$ in a low-dimensional Euclidean space $\mathbb{R}^d$ such that the Euclidean distances $\|\mathbf{z}_i - \mathbf{z}_j\|$ are as close as possible to a given $n \times n$ symmetric, non-negative dissimilarity matrix $\Delta = (\delta_{ij})$ (Torgerson, 1952; Kruskal, 1964a; Borg and Groenen, 2005; Cox and Cox, 2000). The criterion of closeness is made precise by a *stress* function, and the embedding is obtained by minimizing this stress. MDS and its relatives are among the oldest and most widely used tools in multivariate statistics, visualization, and machine learning (Shepard, 1962; Sammon, 1969; van der Maaten and Hinton, 2008; McInnes et al., 2018).

When the input data are given as a matrix $\mathbf{X} \in \mathbb{R}^{n \times m}$ of n observations in a high-dimensional ambient space $\mathbb{R}^m$, the most common choice is $\delta_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_{\mathbb{R}^m}$, the Euclidean distance in the original space. This is appropriate when the data are believed to lie in a linear subspace of $\mathbb{R}^m$, so that Euclidean distances faithfully reflect intrinsic structure. However, when the data lie on or near a nonlinear manifold embedded in $\mathbb{R}^m$, straight-line Euclidean distances are poor proxies for the intrinsic distances that govern the geometry of the data: two points near opposite ends of a curved surface may be close in ambient Euclidean distance but far apart along the surface.

This limitation motivated *Isomap* (Tenenbaum et al., 2000), one of the most influential manifold learning algorithms. Isomap approximates the intrinsic geodesic distance between points by constructing a k -nearest-neighbor (k -NN) graph on the data, computing all-pairs shortest paths on this graph (weighted by Euclidean edge lengths), and then feeding these graph-geodesic dissimilarities $\delta_{ij} = d^{k\text{-NN}}(\mathbf{x}_i, \mathbf{x}_j)$ into classical MDS. For data lying on a smooth, sufficiently sampled Riemannian manifold, the graph-geodesic distances converge to the true manifold geodesics as $n \rightarrow \infty$ (Bernstein et al., 2000; Tenenbaum et al., 2000), and the resulting embedding recovers the intrinsic geometry with high fidelity.

1.1 The Output-Side Inconsistency

Despite this advance, Isomap retains a fundamental asymmetry that has not previously been identified as a source of systematic distortion. On the *input* side, distances are geodesic—shortest paths through the data graph. On the *output* side, the stress function measures faithfulness via *Euclidean* distances between embedded points. This means that Isomap optimizes an embedding in which the distance between $\mathbf{z}_i$ and $\mathbf{z}_j$ is the straight-line distance $\|\mathbf{z}_i - \mathbf{z}_j\|$, not the embedded length of the corresponding input-side geodesic.

The comparison below summarizes the distinction between standard MDS, Isomap, and the proposed Geodesic MDS in terms of the distance used on the input and output sides of the embedding problem.

Method	Input dissimilarity δ_{ij}	Output distance d_{ij}
Standard MDS	Euclidean $\ \cdot\ _{\mathbb{R}^m}$	Euclidean $\ \cdot\ _{\mathbb{R}^d}$
Isomap	d_{G_k} (graph geodesic)	Euclidean $\ \cdot\ _{\mathbb{R}^d}$
Geodesic MDS	d_{G_k} (graph geodesic)	embedded length of a fixed input geodesic

For data living on a smooth manifold with slowly varying curvature this asymmetry causes only a small error: nearby embedded points have approximately equal Euclidean

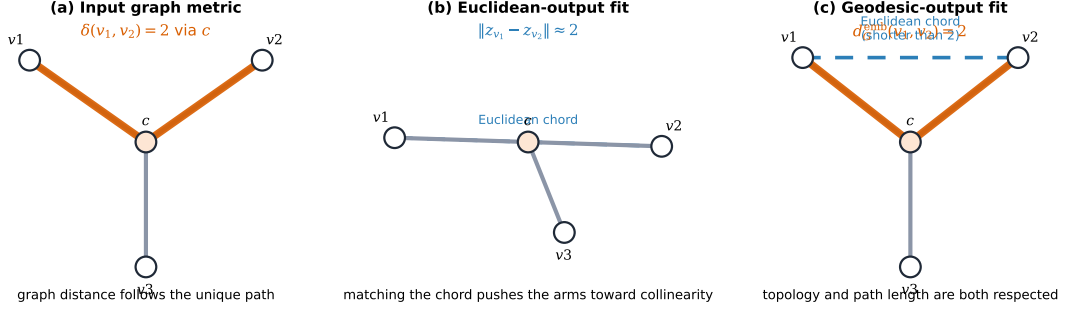


Figure 1: Branching metrics expose the output-side inconsistency. In a star graph, the graph distance between leaves is realized through the center. Matching this distance with Euclidean output distances biases the embedding toward collinearity, whereas measuring output fidelity by the embedded lengths of the corresponding input geodesics preserves the correct topological structure.

and geodesic distances. But for data with genuine graph or tree structure—where the topology enforces that all paths between certain pairs of points must pass through bottleneck vertices—the Euclidean output distance is a systematic *underestimate* of the embedded length of the relevant graph geodesic (by the triangle inequality, with equality only when points are collinear in the embedding space).

Consider a concrete example: a star graph with center c and three arm tips v_1, v_2, v_3 , each arm of unit length. The graph distance between v_1 and v_2 is $\delta(v_1, v_2) = 2$ (through c). Any faithful embedding must respect the topological constraint that the only path from v_1 to v_2 passes through c . Standard MDS and Isomap both try to place $\mathbf{z}_{v_1}$ and $\mathbf{z}_{v_2}$ at Euclidean distance 2 apart, which is perfectly achievable in $\mathbb{R}^2$ —but doing so forces the embedded arms to be nearly collinear, misrepresenting the star topology. The *correct* criterion is that the *path-length distance along the embedded arms* from $\mathbf{z}_{v_1}$ to $\mathbf{z}_{v_2}$ equals $\|\mathbf{z}_{v_1} - \mathbf{z}_c\| + \|\mathbf{z}_c - \mathbf{z}_{v_2}\| = 2$, which is compatible with any angular arrangement of the arms and correctly reflects the star structure. Figure 1 is intended to make this contrast visually explicit.

1.2 Contributions

This paper introduces *Geodesic MDS*, a reformulation of the metric MDS stress function that is self-consistent with respect to geodesic distance. Our contributions are as follows.

1. **Conceptual framework.** We identify and formalize the input/output asymmetry in Isomap and show that replacing Euclidean output distances with embedded lengths of the input geodesics yields a self-consistent stress function (section 3).
2. **Formal definition.** We define the geodesic stress function over a fixed graph topology (the k -NN graph of the input data) and a fixed family of input-side geodesics, where the free variables are the vertex positions in the target space $\mathbb{R}^d$ and the distances are measured along those embedded paths (section 3.3).

3. **Gradient derivation.** We derive the gradient of the geodesic stress with respect to vertex positions and characterize its non-local dependence on the chosen input-side geodesics (section 3.5).
4. **SMACOF extension.** We extend the SMACOF majorization framework (De Leeuw, 1977; De Leeuw and Mair, 2009) to the geodesic setting by exploiting the fact that the fixed input geodesics decompose into sums of Euclidean edge lengths, enabling a tractable iterative optimization (section 2.3).
5. **Geometric characterization.** We characterize the role of k as a geometric scale parameter: $k = 1$ (spanning tree) corresponds to a tree-metric embedding, while $k \rightarrow n$ (complete graph) recovers standard metric MDS (section 3.4). We connect the construction to minimum metric realizations, metric spanners, and the tight span from metric geometry.

1.3 Relationship to Existing Work

Geodesic MDS is closely related to, but distinct from, several existing methods. *Isomap* (Tenenbaum et al., 2000) uses graph geodesics on the input side only. *Kamada-Kawai graph drawing* (Kamada and Kawai, 1989) minimizes a stress between graph-theoretic shortest-path distances and Euclidean drawing distances, but is restricted to 2D visualization and is not formulated as a dimensionality reduction method with explicit input/output distance comparison. *Metric graph embedding* theory (Bourgain, 1985; Linial et al., 1995) studies distortion bounds for embedding graph metrics into ℓ_p spaces but does not define a stress function or optimization criterion. *Diffusion maps* (Coifman and Lafon, 2006) define a Markov-chain-based distance that is intrinsic to the data graph but again uses Euclidean output distances. To our knowledge, the self-consistent formulation proposed here—graph geodesics on the input side and embedded lengths of the corresponding fixed input geodesics on the output side—has not been previously proposed.

1.4 Organization

The remainder of the paper is organized as follows. section 2 reviews MDS, Isomap, and the k -NN graph construction. section 3 presents Geodesic MDS: the formal definition of geodesic stress, the gradient derivation, the SMACOF extension, and the geometric properties of the formulation. Subsequent sections (not included in this draft) present experiments, theoretical analysis, and applications.

2 Background

2.1 Classical MDS

Let $\Delta = (\delta_{ij})$ be an $n \times n$ symmetric dissimilarity matrix with $\delta_{ii} = 0$. Classical MDS (Torgerson, 1952; Young and Householder, 1938) seeks a configuration $\mathbf{Z} \in \mathbb{R}^{n \times d}$ such that $\|\mathbf{z}_i - \mathbf{z}_j\| \approx \delta_{ij}$. When Δ contains squared Euclidean distances, the solution is obtained by eigendecomposition of the *double-centered* Gram matrix

$$\mathbf{B} = -\frac{1}{2}\mathbf{J}\mathbf{D}^{(2)}\mathbf{J}, \quad (1)$$

where $\mathbf{D}_{ij}^{(2)} = \delta_{ij}^2$ and $\mathbf{J} = \mathbf{I}_n - n^{-1}\mathbf{1}\mathbf{1}^\top$ is the centering matrix. The d -dimensional embedding is $\mathbf{Z} = \mathbf{U}_d \Lambda_d^{1/2}$, where $\mathbf{U}_d$ and Λ_d contain the d largest eigenvectors and eigenvalues of $\mathbf{B}$ (Eckart and Young, 1936; Gower, 1966). This is also known as *principal coordinate analysis* (PCoA) (Gower, 1966). Classical MDS is a closed-form, non-iterative method; it yields an exact isometric embedding when $\mathbf{B}$ is positive semidefinite (equivalently, when Δ satisfies Euclidean metric conditions (Schoenberg, 1935)).

2.2 Metric MDS and Stress

When Δ is not a Euclidean distance matrix, classical MDS may place negative eigenvalues in $\mathbf{B}$, and the embedding is approximate. Metric MDS (Kruskal, 1964a) frames the problem explicitly as an optimization: find $\mathbf{Z} \in \mathbb{R}^{n \times d}$ minimizing a *stress* function that measures discrepancy between embedded distances and input dissimilarities.

Kruskal’s Stress-1. The most widely used criterion is Kruskal’s *Stress-1* (Kruskal, 1964a):

$$S_1(\mathbf{Z}) = \sqrt{\frac{\sum_{i < j} (d_{ij}(\mathbf{Z}) - \delta_{ij})^2}{\sum_{i < j} d_{ij}(\mathbf{Z})^2}}, \quad (2)$$

where $d_{ij}(\mathbf{Z}) = \|\mathbf{z}_i - \mathbf{z}_j\|$ is the Euclidean distance between embedded points. The normalization by $\sum_{i < j} d_{ij}^2$ makes S_1 invariant to uniform scaling of $\mathbf{Z}$; it takes values in $[0, 1]$, with 0 indicating a perfect isometric embedding. An alternative normalization divides by $\sum_{i < j} \delta_{ij}^2$ (Borg and Groenen, 2005), and is sometimes called Stress-1r; both are in common use.

Weighted stress. A generalization incorporates pair-specific weights $w_{ij} \geq 0$:

$$\sigma_w(\mathbf{Z}) = \sum_{i < j} w_{ij} (d_{ij}(\mathbf{Z}) - \delta_{ij})^2. \quad (3)$$

Common choices are $w_{ij} = \delta_{ij}^{-2}$ (relative weighting, emphasizing small distances) and $w_{ij} = 1$ (uniform). The unweighted raw stress $\sigma(\mathbf{Z}) = \sum_{i < j} (d_{ij}(\mathbf{Z}) - \delta_{ij})^2$ is the numerator of S_1^2 up to scaling.

Qualitative interpretation. Empirical guidelines due to Kruskal (1964b) and Borg and Groenen (2005) suggest that $S_1 < 0.05$ indicates an excellent fit, 0.05–0.10 a good fit, 0.10–0.20 a fair fit, and > 0.20 a poor fit, though these thresholds depend on n and the structure of Δ .

2.3 SMACOF

Direct minimization of σ or S_1 is a non-convex problem with many local minima. The *Scaling by MAjorizing a COmplicated Function* (SMACOF) algorithm (De Leeuw, 1977; De Leeuw and Mair, 2009) is the dominant approach. SMACOF exploits the identity

$$\sigma(\mathbf{Z}) = \sum_{i < j} w_{ij} \delta_{ij}^2 - 2 \operatorname{tr}(\mathbf{Z}^\top \mathbf{V} \mathbf{Z}) + \operatorname{tr}(\mathbf{Z}^\top \mathbf{W} \mathbf{Z}), \quad (4)$$

where $\mathbf{W}_{ii} = \sum_{j \neq i} w_{ij}$, $\mathbf{W}_{ij} = -w_{ij}$ ($i \neq j$), and $\mathbf{V}$ is the Guttman transform of $\mathbf{Z}$:

$$\mathbf{V}(\mathbf{Z})_{ij} = -w_{ij} \frac{\delta_{ij}}{d_{ij}(\mathbf{Z})} \quad (i \neq j), \quad \mathbf{V}(\mathbf{Z})_{ii} = -\sum_{j \neq i} \mathbf{V}(\mathbf{Z})_{ij}. \quad (5)$$

The majorization step replaces the complicated function of $d_{ij}(\mathbf{Z})$ by a quadratic upper bound that is tight at the current iterate $\mathbf{Z}^{(t)}$, yielding the closed-form Guttman update

$$\mathbf{Z}^{(t+1)} = \mathbf{W}^+ \mathbf{V}(\mathbf{Z}^{(t)}) \mathbf{Z}^{(t)}, \quad (6)$$

where $\mathbf{W}^+$ is the Moore-Penrose pseudoinverse of $\mathbf{W}$. SMACOF guarantees monotone decrease of σ at each iteration and converges to a stationary point (De Leeuw and Mair, 2009). Multiple random restarts are used in practice to mitigate local minima. Efficient implementations are available in R (De Leeuw and Mair, 2009) and Python (Pedregosa et al., 2011).

2.4 Non-Metric MDS

Non-metric MDS (Kruskal, 1964b; Shepard, 1962) relaxes the requirement to reproduce input dissimilarity values, requiring only that the *rank order* of distances be preserved. The optimization alternates between isotonic regression (computing optimal *disparities* $\hat{\delta}_{ij}$ that are monotone in δ_{ij}) and gradient steps on the embedding. Non-metric MDS is appropriate for ordinal or perceptual data; Geodesic MDS as proposed here is a metric method, though a non-metric geodesic variant is straightforward to formulate by applying isotonic regression to the graph-geodesic targets $d_G(i, j)$ while continuing to evaluate the output configuration by the corresponding embedded path lengths $d_G^{\text{emb}}(i, j; \mathbf{Z})$.

2.5 Isomap

Isomap (Tenenbaum et al., 2000) addresses the limitation of MDS on nonlinear manifold-structured data by replacing Euclidean input dissimilarities with approximations of intrinsic geodesic distances. Given $\mathbf{X} \in \mathbb{R}^{n \times m}$, the algorithm proceeds in three steps:

1. **k -NN graph construction.** Build a weighted undirected graph G_k with vertex set $\{1, \dots, n\}$, connecting i and j by an edge of weight $\|\mathbf{x}_i - \mathbf{x}_j\|_{\mathbb{R}^m}$ whenever j is among the k nearest neighbors of i (or vice versa, in the symmetric variant).
2. **Geodesic distance computation.** Compute all-pairs shortest-path distances $d^{k\text{-NN}}(i, j)$ on G_k using Dijkstra's algorithm (Dijkstra, 1959) or Floyd-Warshall (Floyd, 1962). Set $\delta_{ij} := d^{k\text{-NN}}(i, j)$.
3. **Classical MDS.** Apply classical MDS (Section 2.1) to the geodesic dissimilarity matrix Δ to obtain the low-dimensional embedding $\mathbf{Z} \in \mathbb{R}^{n \times d}$.

Convergence theory. For data sampled i.i.d. from a compact Riemannian manifold (M, g) embedded in $\mathbb{R}^m$, the graph-geodesic distance $d^{k\text{-NN}}(i, j)$ converges to the Riemannian geodesic distance $d_M(i, j)$ as $n \rightarrow \infty$, provided k grows appropriately with n (Bernstein et al., 2000; Tenenbaum et al., 2000; Zha and Zhang, 2003; Donoho and Grimes, 2003). The resulting classical MDS embedding then recovers the isometric embedding of M into a Euclidean space (when one exists) (Tenenbaum et al., 2000).

The output-side limitation. Despite this theoretical guarantee, Isomap measures the fidelity of the embedding by Euclidean distances $\|\mathbf{z}_i - \mathbf{z}_j\|$ between embedded points—not by path-length distances along the embedded manifold. For a smooth manifold this is a minor issue: the manifold in the embedding is nearly flat if d is chosen correctly, so Euclidean and geodesic distances are approximately equal. For discrete graph metrics, branching metrics (trees, stars), or data with topological bottlenecks, the discrepancy is fundamental, as we elaborate in section 3.

2.6 The k -NN Graph as a Geodesic Structure

The k -NN graph G_k constructed in step 1 of Isomap can be understood as a *metric graph*—a weighted graph whose path metric approximates some underlying geodesic metric space. Several observations are relevant for our construction.

Consistency with the ambient metric. The edge weights of G_k are Euclidean distances in $\mathbb{R}^m$, so the path metric d_{G_k} always satisfies $d_{G_k}(i, j) \geq \|\mathbf{x}_i - \mathbf{x}_j\|$ (the shortest path cannot be shorter than the straight line). Equality holds only for pairs connected by a direct edge.

Sparsity and the role of k . G_k has at most kn edges (exactly $kn/2$ in expectation for large n with a symmetric construction), so the graph is sparse for small k . As k increases, more edges are added, $d_{G_k}(i, j)$ decreases toward $\|\mathbf{x}_i - \mathbf{x}_j\|$, and the graph metric approaches the Euclidean ambient metric. At $k = n - 1$ (complete graph), $d_{G_k}(i, j) = \|\mathbf{x}_i - \mathbf{x}_j\|$ for all pairs, and the construction degenerates to standard Euclidean MDS.

Spanner interpretation. G_k is a $(1 + \varepsilon)$ -spanner of the Euclidean metric on $\mathbf{X}$ for sufficiently large k depending on the intrinsic dimensionality of the data (Althöfer et al., 1993; Har-Peled and Mendel, 2006). This means $d_{G_k}(i, j) \leq (1 + \varepsilon) \|\mathbf{x}_i - \mathbf{x}_j\|$, so the graph geodesic approximates the Euclidean distance with controlled relative error. For data with low intrinsic dimension d^* , $k = O(\varepsilon^{-d^*})$ suffices (Har-Peled and Mendel, 2006).

Minimum realization. An edge (i, j) in K_n (the complete graph on $\mathbf{X}$) is *metrically redundant* with respect to d_{G_k} if there exists a path through intermediate points with equal total length. The subgraph obtained by removing all metrically redundant edges is the *minimum realization* G^* of the metric d_{G_k} (Dress, 1984; Imrich et al., 1984), and encodes the intrinsic geodesic structure of the data. For tree metrics, G^* is the unique spanning tree; for hyperbolic metrics it is sparse; for generic metrics it approaches the complete graph. Figure 2 is a natural companion here because the role of k is one of the main geometric ideas of the paper.

2.7 Manifold Learning and Related Methods

The broader context of our work is *manifold learning*: the project of finding low-dimensional representations of high-dimensional data assumed to lie near a low-dimensional manifold. Beyond Isomap, influential methods include Locally Linear Embedding (LLE) (Roweis and Saul, 2000), Laplacian Eigenmaps (Belkin and Niyogi, 2003), Hessian Eigenmaps (Donoho and Grimes, 2003), Diffusion Maps (Coifman and Lafon, 2006), t -SNE (van der Maaten and Hinton, 2008), and UMAP (McInnes et al., 2018). With the exception of diffusion maps

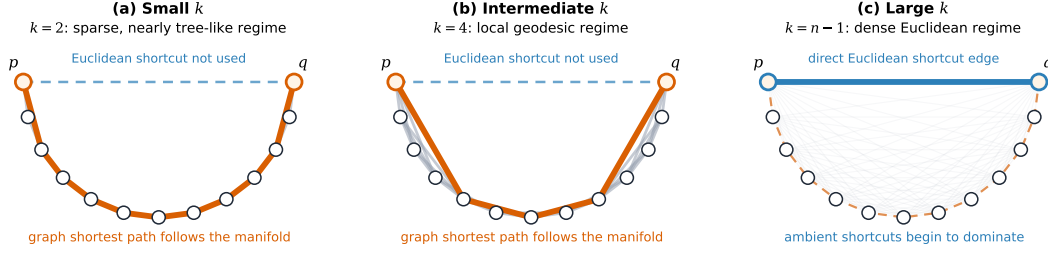


Figure 2: The neighborhood parameter k acts as a geometric scale parameter. The same horseshoe-shaped point set is shown for three neighborhood sizes, with the same endpoint pair p, q highlighted in each panel. For small k , the graph is sparse and nearly tree-like; for intermediate k , shortest paths still follow the manifold and preserve local geodesic structure; and for large k , dense shortcut edges appear and the metric approaches the ambient Euclidean regime.

(which define an intrinsic distance via Markov chain commute times), all of these methods use Euclidean distances on the output side, making the inconsistency identified here relevant to the entire manifold learning literature.

Graph drawing methods—particularly Kamada-Kawai (Kamada and Kawai, 1989) and Fruchterman-Reingold (Fruchterman and Reingold, 1991)—are also related. Kamada-Kawai minimizes a stress between graph-theoretic shortest-path distances (in the unweighted graph) and Euclidean distances in the 2D drawing, which is structurally similar to Isomap applied to an unweighted graph. A geodesic variant of Kamada-Kawai, using embedded lengths of fixed graph geodesics on the output, is a natural companion to Geodesic MDS but is outside the scope of the present paper.

3 Geodesic MDS

3.1 The Self-Consistency Principle

We begin by articulating the principle that motivates the proposed method.

Definition 1 (Self-consistent embedding) *Let (X, δ) be a metric space with a graph structure $G = (X, E, w)$ whose path metric d_G satisfies $d_G(i, j) \geq \delta(i, j)$ for all i, j . For each pair (i, j) , choose a canonical d_G -geodesic γ_{ij} in G . An embedding $f : X \rightarrow \mathbb{R}^d$ is self-consistent with respect to (G, Γ) , where $\Gamma = \{\gamma_{ij}\}_{i < j}$, if the fidelity of the embedding is measured by the path length of the image $f(\gamma_{ij})$ —i.e., by the sum of Euclidean edge lengths along that same input-side geodesic.*

Standard metric MDS and Isomap are not self-consistent in this sense: they measure fidelity by Euclidean distances $\|f(i) - f(j)\|$, which are at most equal to any path-length distance through intermediate vertices.

The self-consistency principle has a natural interpretation: *the distance used to evaluate the embedding should be the same type of distance, along the same family of geodesics, used*

to construct the input dissimilarities. If input dissimilarities are graph geodesics, output distances should be measured along those same graph geodesics after embedding. This is not merely a philosophical preference: the analysis in section 3.2 below shows that using Euclidean output distances introduces a systematic downward bias in the estimated distances for topologically constrained pairs.

3.2 Systematic Bias of Euclidean Output Distances

Proposition 2 (Triangle inequality bias) *Let $\mathbf{Z} = (\mathbf{z}_1, \dots, \mathbf{z}_n) \in \mathbb{R}^{n \times d}$ be any embedding, and let $G = (V, E, w)$ be a weighted graph with vertex set $V = \{1, \dots, n\}$. For any pair (i, j) with chosen shortest path $i = v_0, v_1, \dots, v_\ell = j$ in G , define the embedded length of that geodesic by*

$$d_G^{\text{emb}}(i, j; \mathbf{Z}) = \sum_{s=0}^{\ell-1} \|\mathbf{z}_{v_s} - \mathbf{z}_{v_{s+1}}\|. \quad (7)$$

Then

$$\|\mathbf{z}_i - \mathbf{z}_j\| \leq d_G^{\text{emb}}(i, j; \mathbf{Z}), \quad (8)$$

with equality if and only if all embedded path vertices $\mathbf{z}_{v_0}, \dots, \mathbf{z}_{v_\ell}$ are collinear in $\mathbb{R}^d$ in the order of the path.

Proof By the triangle inequality applied ℓ times: $\|\mathbf{z}_i - \mathbf{z}_j\| = \|\sum_s (\mathbf{z}_{v_{s+1}} - \mathbf{z}_{v_s})\| \leq \sum_s \|\mathbf{z}_{v_{s+1}} - \mathbf{z}_{v_s}\| = d_G^{\text{emb}}(i, j; \mathbf{Z})$. Equality holds iff all difference vectors $\mathbf{z}_{v_{s+1}} - \mathbf{z}_{v_s}$ are non-negative scalar multiples of a single unit vector, i.e., the points are collinear in the stated order. $\blacksquare$

Theorem 2 shows that standard MDS and Isomap systematically underestimate embedded geodesic lengths for all topologically non-degenerate pairs (those whose shortest path has $\ell \geq 2$ and does not embed as a straight line). When the input dissimilarity δ_{ij} is a graph geodesic, minimizing Euclidean-output stress creates pressure to satisfy $\|\mathbf{z}_i - \mathbf{z}_j\| \approx \delta_{ij}$, which in turn forces embedded paths to be approximately collinear. For tree-structured data this collapses the tree into a lower-dimensional structure than warranted; for star graphs it forces nearly collinear arm arrangement regardless of the dimension d .

3.3 The Geodesic Stress Function

We now define the proposed method.

Setup. Let $G = (V, E, w)$ be a connected weighted graph with vertex set $V = \{1, \dots, n\}$ and positive edge weights $w_{uv} > 0$. Write d_G for the weighted shortest-path metric on V . For each pair $i < j$, choose a canonical d_G -geodesic

$$\gamma_{ij} = (v_0 = i, v_1, \dots, v_L = j), \quad (9)$$

breaking ties deterministically when multiple shortest paths exist. The collection $\Gamma = \{\gamma_{ij} : i < j\}$ is part of the data of the method. In this sense, Geodesic MDS is defined on the pair (G, Γ) , not on G alone.

An embedding is a configuration matrix $\mathbf{Z} = (\mathbf{z}_1, \dots, \mathbf{z}_n)^\top \in \mathbb{R}^{n \times d}$. For fixed (G, Γ) , the *embedded graph* $G(\mathbf{Z})$ is the same graph G with edge lengths replaced by embedded Euclidean distances. The *embedded length of the chosen input geodesic* is

$$d_\Gamma^{\text{emb}}(i, j; \mathbf{Z}) = \sum_{s=0}^{L-1} \|\mathbf{z}_{v_s} - \mathbf{z}_{v_{s+1}}\|, \quad (10)$$

where $\gamma_{ij} = (v_0, \dots, v_L)$.

Definition 3 (Raw geodesic stress) *The raw geodesic stress of an embedding $\mathbf{Z} \in \mathbb{R}^{n \times d}$ with respect to (G, Γ) is*

$$\text{Stress}_\Gamma^{\text{geo}}(\mathbf{Z}; G) = \sum_{i < j} (d_\Gamma^{\text{emb}}(i, j; \mathbf{Z}) - d_G(i, j))^2. \quad (11)$$

Definition 4 (Geodesic Stress-1) *The normalized geodesic Stress-1 of $\mathbf{Z}$ with respect to (G, Γ) is*

$$S_{1\Gamma}^{\text{geo}}(\mathbf{Z}; G) = \sqrt{\frac{\text{Stress}_\Gamma^{\text{geo}}(\mathbf{Z}; G)}{\sum_{i < j} d_G(i, j)^2}}. \quad (12)$$

Definition 5 (Geodesic MDS embedding) *A Geodesic MDS embedding of (G, Γ) in $\mathbb{R}^d$ is any minimizer*

$$\hat{\mathbf{Z}}_{\Gamma, d}(G) \in \arg \min_{\mathbf{Z} \in \mathbb{R}^{n \times d}} \text{Stress}_\Gamma^{\text{geo}}(\mathbf{Z}; G). \quad (13)$$

Equivalently, since the denominator in eq. (12) is constant in $\mathbf{Z}$, one may minimize $S_{1\Gamma}^{\text{geo}}(\mathbf{Z}; G)$.

Specialization to k -NN graphs. When the input is a data matrix $\mathbf{X} \in \mathbb{R}^{n \times m}$, we recover the construction used throughout this paper by fixing a neighborhood parameter $k \in \{1, \dots, n-1\}$ and building the symmetric k -NN graph $G_k = (V, E_k)$ with

$$E_k = \{(i, j) : j \in \mathcal{N}_k(i) \text{ or } i \in \mathcal{N}_k(j)\}, \quad (14)$$

where $\mathcal{N}_k(i)$ denotes the k nearest neighbors of i in $\mathbb{R}^m$. Assign weights $w_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_{\mathbb{R}^m}$ for $(i, j) \in E_k$, compute the shortest-path metric $d_{G_k}(i, j) = \delta_{ij}^{(k)}$, and choose a canonical shortest-path family $\Gamma_k = \{\gamma_{ij}^{(k)} : i < j\}$. The resulting Geodesic MDS objective is

$$\text{Stress}_{\Gamma_k}^{\text{geo}}(\mathbf{Z}; G_k) = \sum_{i < j} (d_{\Gamma_k}^{\text{emb}}(i, j; \mathbf{Z}) - \delta_{ij}^{(k)})^2. \quad (15)$$

Remark 6 (Normalization) *We normalize by the constant $\sum_{i < j} d_G(i, j)^2$ rather than by an embedded denominator such as $\sum_{i < j} d_\Gamma^{\text{emb}}(i, j; \mathbf{Z})^2$. This keeps $S_{1\Gamma}^{\text{geo}}$ scale-free while leaving the optimizer identical to the raw numerator eq. (11). A Kruskal-style denominator depending on $\mathbf{Z}$ is still useful for reporting, but it changes the optimization target.*

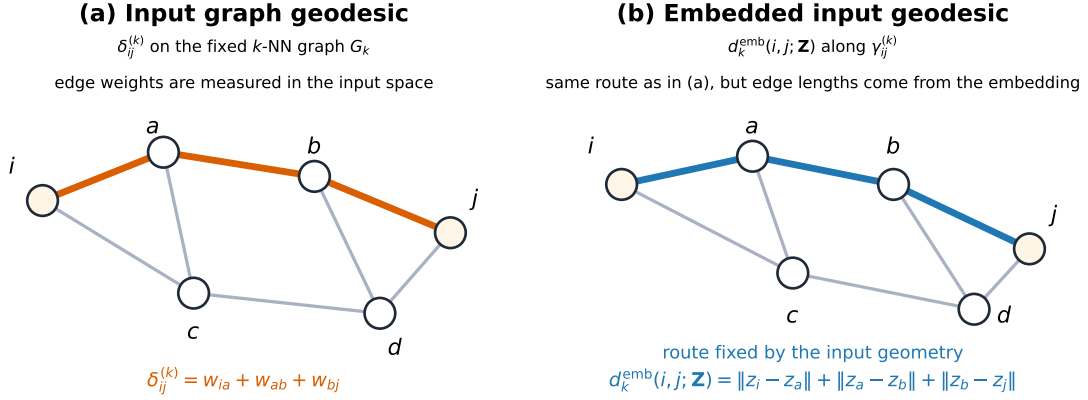


Figure 3: Definition of geodesic stress. Left: the input dissimilarity $d_G(i, j)$ is the shortest-path distance between i and j on the fixed weighted graph G ; in the data-analytic specialization, $G = G_k$ is a k -NN graph. Right: the output quantity $d_{\Gamma}^{\text{emb}}(i, j; \mathbf{Z})$ is the length of the canonical shortest path chosen in the input graph, measured after the vertex positions move in $\mathbb{R}^d$. The route is fixed by the input geometry; only the embedded edge lengths change during optimization.

Remark 7 (Fixed topology and fixed geodesics) *The graph topology E and the chosen geodesic family Γ are fixed by the input metric data and are not part of the optimization. Only the vertex positions $\mathbf{z}_1, \dots, \mathbf{z}_n$ are free. This is a key design choice: it ensures that the relevant routes are determined by the input geometry and that the optimization preserves that chosen geodesic structure after embedding.*

Figure 3 should make the fixed-geodesic, two-metric comparison in eqs. (10) and (12) easy to see.

Remark 8 (Nonuniqueness is intrinsic, not a defect) *The objective eq. (11) is a fidelity criterion, not an aesthetics criterion, so its minimizers need not be unique. This is a feature of the definition: multiple embeddings can realize the same graph geodesic data equally well.*

For example, for a path graph $v_0 - v_1 - v_2$ with any fixed edge lengths, every embedding in $\mathbb{R}^2$ that preserves those two edge lengths is a global minimum of eq. (11); one obtains an infinite family of minima by fixing the segment $[v_0, v_1]$ and rotating $[v_1, v_2]$ around v_1 . Likewise, if v_0, v_1, v_2 form one triangle and v_1, v_2, v_3 form another triangle sharing the edge $[v_1, v_2]$, then both the non-overlapping and the overlapping realizations are global minima whenever all edge lengths are preserved. Figure 4 illustrates these two basic sources of nonuniqueness.

A diagnostic lesson. The diagnostic experiments reported separately in this project show that the all-pairs objective eq. (11) can admit visually pathological minimizers on simple

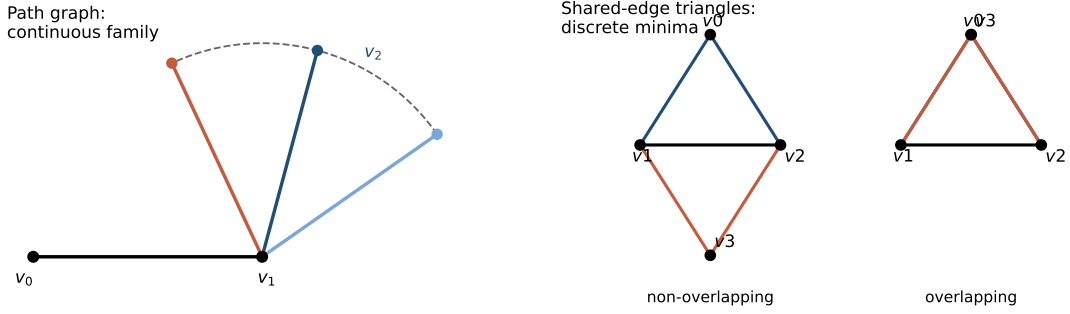


Figure 4: Two sources of nonuniqueness for geodesic-stress minimizers. Left: a three-vertex path has a one-parameter family of exact realizations obtained by rotating the second edge while preserving both edge lengths. Right: two triangles sharing an edge admit distinct exact realizations, including a non-overlapping embedding and an overlapping embedding. In both examples the raw geodesic stress is zero because all relevant graph-geodesic lengths are preserved.

synthetic graphs. On orthogonal grids, graph geodesics systematically exceed straight-line Euclidean distances, so the gradient of the all-pairs geodesic stress can push the embedding toward stretched or folded configurations. This does not invalidate the definition of GMDS: it shows that regularization is a separate modeling choice that may be needed when one wants both fidelity to graph geodesics and a visually well-spread realization.

Regularized GMDS. To incorporate such modeling choices explicitly, introduce a Euclidean dispersion regularizer

$$R(\mathbf{Z}) = \sum_{i < j} f_r(\|\mathbf{z}_i - \mathbf{z}_j\|), \quad (16)$$

where $f_r : (0, \infty) \rightarrow (0, \infty)$ is a decreasing function. A concrete choice used widely in force-directed graph drawing is the Coulomb-type penalty

$$f_r(r) = \frac{1}{r^2 + \varepsilon_r^2}, \quad (17)$$

with a small $\varepsilon_r > 0$ to prevent singularities. The self-consistency principle applies to the fidelity term eq. (11); the role of R is different. It is a Euclidean regularizer that encourages spread and discourages collapse, not a substitute for the graph-geodesic mismatch term itself. A standard scalarized objective is therefore

$$F_\lambda(\mathbf{Z}; G, \Gamma) = \text{Stress}_\Gamma^{\text{geo}}(\mathbf{Z}; G) + \lambda R(\mathbf{Z}), \quad \lambda > 0. \quad (18)$$

A *regularized GMDS embedding* is a local minimizer of eq. (18) for a specified λ . The value of λ is not universal; in practice it should be reported explicitly and chosen by a fidelity-dispersion trade-off study, continuation scheme, or task-specific validation criterion.

An edge-based relaxation. If exact minimization of eq. (11) is computationally prohibitive, one can replace it by the edge-only functional

$$E_K(\mathbf{Z}) = \sum_{(i,j) \in E} k_{ij} (\|\mathbf{z}_i - \mathbf{z}_j\| - w_{ij})^2, \quad (19)$$

because for an edge (i, j) the chosen graph geodesic is the edge itself and thus $d_{\Gamma}^{\text{emb}}(i, j; \mathbf{Z}) = \|\mathbf{z}_i - \mathbf{z}_j\|$. This is no longer the geodesic stress: it is a graph-drawing-style relaxation that penalizes only single-hop distortion, and is therefore closer to an edge-restricted Kamada-Kawai objective (Kamada and Kawai, 1989; Fruchterman and Reingold, 1991) than to the full all-pairs GMDS criterion.

The relation between eqs. (11) and (19) is easiest to see in edge-length coordinates. Let $\ell(\mathbf{Z}) \in \mathbb{R}^{|E|}$ be the vector of embedded edge lengths, let $w \in \mathbb{R}^{|E|}$ collect the graph edge weights, and let A_{Γ} be the path-edge incidence matrix whose (i, j) -row indicates which edges lie on the chosen geodesic γ_{ij} . Then

$$\text{Stress}_{\Gamma}^{\text{geo}}(\mathbf{Z}; G) = \|A_{\Gamma}(\ell(\mathbf{Z}) - w)\|_2^2 = (\ell(\mathbf{Z}) - w)^{\top} A_{\Gamma}^{\top} A_{\Gamma} (\ell(\mathbf{Z}) - w), \quad (20)$$

whereas

$$E_K(\mathbf{Z}) = (\ell(\mathbf{Z}) - w)^{\top} K (\ell(\mathbf{Z}) - w), \quad (21)$$

with $K = \text{diag}(k_{ij})$ diagonal. Thus E_K drops the off-diagonal path-coupling terms in $A_{\Gamma}^{\top} A_{\Gamma}$ and therefore does not control multi-hop geodesic distortion directly. What is gained is a much cheaper objective; what is lost is explicit control of long-range path-length errors.

If $E_K(\mathbf{Z}) = 0$, then every edge length is realized exactly, hence every chosen graph-geodesic length is realized exactly as well, and eq. (11) also vanishes. In the approximate regime, however, the two objectives are different and generally have different local minima. For this reason we treat eq. (19) as a relaxation of GMDS, not as the definition of GMDS itself. Whether this relaxation also avoids some of the pathologies of the full all-pairs objective is an empirical question, not a formal consequence of the definition.

The weights k_{ij} in eq. (19) are user-chosen tuning parameters. A simple default is $k_{ij} \equiv 1$. If one wants to emphasize edge lengths that occur frequently in the graph, one may set $k_{ij} = \hat{f}_{\text{edge}}(w_{ij})$, where $\hat{f}_{\text{edge}}$ is a kernel density estimate of the empirical distribution of edge lengths. Other choices, such as weights proportional to geodesic-usage counts, are also possible and should be justified by the application.

3.4 Properties of Geodesic MDS

Self-consistency. Geodesic MDS satisfies theorem 1 by construction: both input dissimilarities and output distances are measured on the same fixed family of geodesics in (G, Γ) (the former by the graph metric d_G , the latter by the embedded path lengths $d_{\Gamma}^{\text{emb}}(i, j; \mathbf{Z})$).

Recovery of standard MDS at $k = n - 1$. When the input graph is the k -NN specialization and $k = n - 1$, every pair is connected by a direct edge, so $d_{\Gamma_{n-1}}^{\text{emb}}(i, j; \mathbf{Z}) = \|\mathbf{z}_i - \mathbf{z}_j\|$ and $\delta_{ij}^{(n-1)} = \|\mathbf{x}_i - \mathbf{x}_j\|$. Geodesic stress reduces to standard metric Stress-1, and Geodesic MDS recovers standard metric MDS.

Tree metric at small k . For $k = 1$, G_k is (generically) a sparse forest; with a suitable connectivity correction (e.g., taking the minimum spanning tree as G_1) the graph is a tree. The tree-path metric is then a tree metric in the sense of Dress (1984), and Geodesic MDS seeks an embedding that faithfully represents the tree structure. Tree metrics embed isometrically into $\mathbb{R}^1$ (the real line) with zero stress; thus for tree-structured data Geodesic MDS with $k = 1$ achieves perfect stress in $d = 1$.

Interpolation and the role of k . As k increases from 1 to $n - 1$, more edges are added to G_k , $\delta_{ij}^{(k)}$ decreases toward $\|\mathbf{x}_i - \mathbf{x}_j\|$, and $d_{\Gamma_k}^{\text{emb}}(i, j; \mathbf{Z})$ decreases toward $\|\mathbf{z}_i - \mathbf{z}_j\|$. The geodesic stress thus interpolates between a tree-metric embedding at small k and standard MDS at large k . The parameter k controls the geometric scale at which geodesic structure is enforced, analogous to the bandwidth parameter in kernel methods.

Geodesic stress is never smaller than Euclidean stress. By theorem 2, $d_{\Gamma_k}^{\text{emb}}(i, j; \mathbf{Z}) \geq \|\mathbf{z}_i - \mathbf{z}_j\|$ for all i, j and all $\mathbf{Z}$. If in addition $\delta_{ij}^{(k)} \geq \|\mathbf{x}_i - \mathbf{x}_j\|$ (which holds by the triangle inequality on graph paths), then for the same configuration $\mathbf{Z}$ the geodesic stress is larger than the Euclidean stress. The optimal geodesic configuration $\hat{\mathbf{Z}}_{\Gamma_k, d}(G_k)$ is not in general the same as the optimal Euclidean configuration.

3.5 Gradient Derivation

We derive the gradient of the raw geodesic stress $\text{Stress}_{\Gamma_k}^{\text{geo}}(\mathbf{Z}; G_k)$ (eq. (15)) with respect to the position $\mathbf{z}_\ell \in \mathbb{R}^d$ of a single vertex ℓ . For brevity, write $\sigma^{\text{geo}}(\mathbf{Z}) = \text{Stress}_{\Gamma_k}^{\text{geo}}(\mathbf{Z}; G_k)$.

Fixed geodesics and differentiability. For each pair $i < j$, let $\gamma_{ij}^{(k)} = (v_0 = i, v_1, \dots, v_L = j)$ be the canonical shortest path chosen once from the input graph G_k . Then $d_{\Gamma_k}^{\text{emb}}(i, j; \mathbf{Z})$ is the length of this fixed polygonal chain in the embedding. Since the route does not change with $\mathbf{Z}$, the only nonsmoothness comes from Euclidean norms at zero-length edges; away from such degenerate configurations, $d_{\Gamma_k}^{\text{emb}}$ is differentiable.

Gradient with respect to an intermediate vertex. Let $\gamma_{ij}^{(k)} = (v_0, v_1, \dots, v_L)$ be the chosen shortest path from $i = v_0$ to $j = v_L$. Define the unit edge vector

$$\mathbf{e}_{ab} = \frac{\mathbf{z}_a - \mathbf{z}_b}{\|\mathbf{z}_a - \mathbf{z}_b\|}, \quad (22)$$

so that $\frac{\partial}{\partial \mathbf{z}_a} \|\mathbf{z}_a - \mathbf{z}_b\| = \mathbf{e}_{ab}$ and $\frac{\partial}{\partial \mathbf{z}_b} \|\mathbf{z}_a - \mathbf{z}_b\| = -\mathbf{e}_{ab}$.

The gradient of $d_{\Gamma_k}^{\text{emb}}(i, j; \mathbf{Z})$ with respect to $\mathbf{z}_\ell$ depends on whether ℓ appears in $\gamma_{ij}^{(k)}$:

$$\frac{\partial d_{\Gamma_k}^{\text{emb}}(i, j; \mathbf{Z})}{\partial \mathbf{z}_\ell} = \begin{cases} \mathbf{e}_{\ell, v_{s-1}} + \mathbf{e}_{\ell, v_{s+1}} & \text{if } \ell = v_s, \ 0 < s < L, \\ \mathbf{e}_{\ell, v_1} & \text{if } \ell = v_0 = i, \\ \mathbf{e}_{\ell, v_{L-1}} & \text{if } \ell = v_L = j, \\ \mathbf{0} & \text{if } \ell \notin \gamma_{ij}^{(k)}. \end{cases} \quad (23)$$

The formula captures the non-local nature of geodesic MDS: only those pairs (i, j) whose chosen input geodesic passes through vertex ℓ contribute to the gradient at $\mathbf{z}_\ell$. Figure 5 should help readers see why the gradient depends on global geodesic-route structure rather than only on immediate graph neighbors.

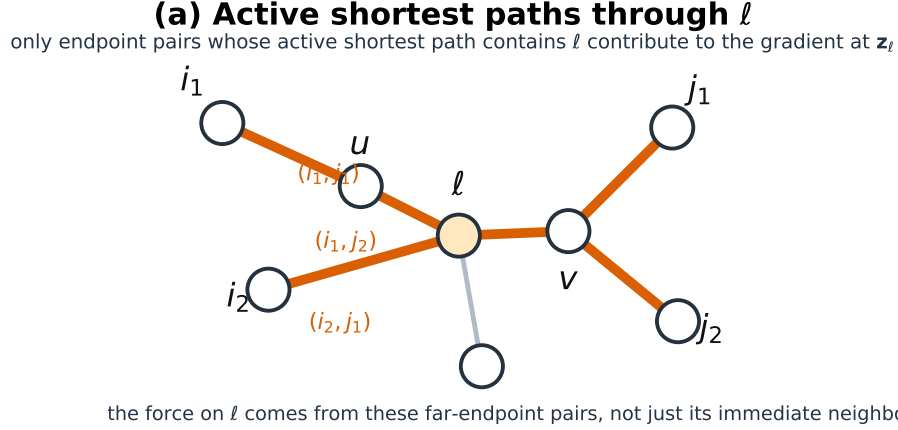


Figure 5: Geodesic stress has non-local gradients. A vertex receives contributions only from endpoint pairs whose chosen input geodesic passes through it, so the gradient depends on global route structure rather than only on immediate graph neighbors.

Gradient of the raw geodesic stress. By the chain rule,

$$\frac{\partial \sigma^{\text{geo}}}{\partial \mathbf{z}_\ell} = 2 \sum_{i < j} (d_{\Gamma_k}^{\text{emb}}(i, j; \mathbf{Z}) - \delta_{ij}^{(k)}) \cdot \frac{\partial d_{\Gamma_k}^{\text{emb}}(i, j; \mathbf{Z})}{\partial \mathbf{z}_\ell}. \quad (24)$$

Substituting eq. (23) and collecting terms, the gradient decomposes into contributions from paths that pass through ℓ as an *internal* vertex vs. paths that have ℓ as an *endpoint*:

$$\begin{aligned} \frac{\partial \sigma^{\text{geo}}}{\partial \mathbf{z}_\ell} = & 2 \sum_{\substack{i < j \\ \ell \in \gamma_{ij}^{(k)}, \ell \neq i, j}} r_{ij} (\mathbf{e}_{\ell, \text{prev}(\gamma_{ij}^{(k)})} + \mathbf{e}_{\ell, \text{next}(\gamma_{ij}^{(k)})}) \\ & + 2 \sum_{\substack{j > \ell \\ \ell \in \gamma_{\ell j}^{(k)}}} r_{\ell j} \mathbf{e}_{\ell, \text{next}(\gamma_{\ell j}^{(k)})} + 2 \sum_{\substack{i < \ell \\ \ell \in \gamma_{i\ell}^{(k)}}} r_{i\ell} \mathbf{e}_{\ell, \text{prev}(\gamma_{i\ell}^{(k)})}, \end{aligned} \quad (25)$$

where $r_{ij} = d_{\Gamma_k}^{\text{emb}}(i, j; \mathbf{Z}) - \delta_{ij}^{(k)}$ is the signed residual and $\text{prev}(\gamma_{ij}^{(k)})$ (resp. $\text{next}(\gamma_{ij}^{(k)})$) denotes the predecessor (resp. successor) of ℓ on the fixed path $\gamma_{ij}^{(k)}$. The entire gradient matrix $\nabla_{\mathbf{Z}} \sigma^{\text{geo}} \in \mathbb{R}^{n \times d}$ can be assembled in $O(n^2 L_{\max})$ operations, where $L_{\max}$ is the maximum path length in G_k .

Degenerate edge lengths. The geodesic stress is not differentiable only when some embedded edge on a chosen geodesic collapses to zero length. At such a point, one may use the standard subgradient of the Euclidean norm or introduce a small edge-length stabilizer. Away from these measure-zero degeneracies, the gradient above is valid.

3.6 SMACOF Majorization for Geodesic Stress

The standard SMACOF majorization exploits the representation $d_{ij}(\mathbf{Z}) = \|\mathbf{z}_i - \mathbf{z}_j\|$ as a single Euclidean norm, which can be bounded below by a linear function of $\mathbf{Z}$ via the Cauchy-Schwarz inequality. For geodesic MDS, $d_{\Gamma_k}^{\text{emb}}(i, j; \mathbf{Z})$ is a *sum* of Euclidean norms along a path. This structure still admits a majorization, as we now show.

Fixed-geodesic majorization. Fix a reference configuration $\mathbf{Z}^{(t)}$. The path family Γ_k is already fixed by the input metric, so $d_{\Gamma_k}^{\text{emb}}(i, j; \mathbf{Z})$ is always a sum of norms along the stored path $\gamma_{ij}^{(k)}$. Each individual norm $\|\mathbf{z}_a - \mathbf{z}_b\|$ admits the Cauchy-Schwarz majorization

$$\|\mathbf{z}_a - \mathbf{z}_b\| \leq \frac{\|\mathbf{z}_a - \mathbf{z}_b\|^2 + \|\mathbf{z}_a^{(t)} - \mathbf{z}_b^{(t)}\|^2}{2 \|\mathbf{z}_a^{(t)} - \mathbf{z}_b^{(t)}\|} \quad (\text{tight at } \mathbf{Z} = \mathbf{Z}^{(t)}). \quad (26)$$

Summing over path edges, the total geodesic distance is majorized by a function that is quadratic in $\mathbf{Z}$, and hence the stress is majorized by a fourth-degree polynomial. A tighter approach uses the standard SMACOF identity applied path-edge by path-edge.

Path-augmented Guttman transform. Define the *path-edge weight* $\tilde{w}_{ab}^{(t)}$ as the total weight accumulated by edge (a, b) across all chosen input geodesics:

$$\tilde{w}_{ab}^{(t)} = \sum_{i < j: (a,b) \in \gamma_{ij}^{(k)}} \frac{1}{\|\mathbf{z}_a^{(t)} - \mathbf{z}_b^{(t)}\|}, \quad (27)$$

and the *path-augmented Guttman matrix*

$$\tilde{\mathbf{V}}_{ab}^{(t)} = -\tilde{w}_{ab}^{(t)}, \quad a \neq b; \quad \tilde{\mathbf{V}}_{aa}^{(t)} = \sum_{b \neq a} \tilde{w}_{ab}^{(t)}. \quad (28)$$

The majorization step then takes the form

$$\mathbf{Z}^{(t+1)} = (\tilde{\mathbf{W}}^{(t)})^+ \tilde{\mathbf{V}}^{(t)} \mathbf{Z}^{(t)}, \quad (29)$$

where $\tilde{\mathbf{W}}^{(t)}$ is the diagonal matrix of row sums of $|\tilde{\mathbf{V}}^{(t)}|$. Since the geodesic family is fixed, the stored paths are reused at every iteration and only the embedded edge lengths are updated.

Proposition 9 (Monotone decrease) *With the geodesic family Γ_k fixed by the input metric, each step (29) does not increase $\sigma^{\text{geo}}(\mathbf{Z})$.*

Proof [Proof sketch] At iteration t , the fixed-geodesic majorization gives an upper bound $\sigma^{\text{geo}}(\mathbf{Z}) \leq f^{(t)}(\mathbf{Z})$ for a function $f^{(t)}$ that is quadratic in $\mathbf{Z}$ and tight at $\mathbf{Z}^{(t)}$. The update (29) minimizes $f^{(t)}$, so $f^{(t)}(\mathbf{Z}^{(t+1)}) \leq f^{(t)}(\mathbf{Z}^{(t)}) = \sigma^{\text{geo}}(\mathbf{Z}^{(t)})$. Because the same fixed path family is used in both the objective and the majorizer, we also have $\sigma^{\text{geo}}(\mathbf{Z}^{(t+1)}) \leq f^{(t)}(\mathbf{Z}^{(t+1)})$, hence

$$\sigma^{\text{geo}}(\mathbf{Z}^{(t+1)}) \leq f^{(t)}(\mathbf{Z}^{(t+1)}) \leq f^{(t)}(\mathbf{Z}^{(t)}) = \sigma^{\text{geo}}(\mathbf{Z}^{(t)}).$$

■

Computational complexity. Each iteration of the geodesic SMACOF requires: (i) path accumulation for $\tilde{\mathbf{W}}^{(t)}$ and $\tilde{\mathbf{V}}^{(t)}$ using the stored geodesic family Γ_k : $O(n^2 L_{\max})$; (ii) a linear system solve: $O(n^3)$ or $O(n^2 d)$ with conjugate gradient. The dominant cost is $O(n^3)$, the same as standard SMACOF. The all-pairs shortest paths and the canonical geodesic family are computed once in preprocessing at cost $O(n(|E_k| + n \log n))$ using Dijkstra with a Fibonacci heap (Fredman and Tarjan, 1987). Figure 6 summarizes the method at the level of data flow and is intended to complement the pseudocode in algorithm 1.

3.7 Algorithm

We summarize Geodesic MDS as algorithm 1.

Algorithm 1 Geodesic MDS

Require: Data $\mathbf{X} \in \mathbb{R}^{n \times m}$, target dimension d , neighborhood size k , convergence tolerance $\varepsilon > 0$, maximum iterations T .

Ensure: Embedding $\mathbf{Z} \in \mathbb{R}^{n \times d}$.

// Step 1: Build input graph geodesic

- 1: Construct symmetric k -NN graph G_k on $\mathbf{X}$ with Euclidean edge weights.
- 2: Compute all-pairs shortest paths $\delta_{ij}^{(k)}$ on G_k (Dijkstra / Floyd-Warshall).
- 3: Choose and store a canonical shortest path family $\Gamma_k = \{\gamma_{ij}^{(k)} : i < j\}$ on G_k .

// Step 2: Initialize

- 4: Initialize $\mathbf{Z}^{(0)}$ via classical MDS on $\Delta^{(k)}$.
- 5: $t \leftarrow 0$.

// Step 3: Geodesic SMACOF iterations

6: **repeat**

7: Evaluate the embedded path lengths $d_{\Gamma_k}^{\text{emb}}(i, j; \mathbf{Z}^{(t)})$ along the stored geodesics $\gamma_{ij}^{(k)}$.

8: Accumulate path-augmented Guttman matrices $\tilde{\mathbf{V}}^{(t)}, \tilde{\mathbf{W}}^{(t)}$ via eqs. (27) and (28).

9: Update $\mathbf{Z}^{(t+1)} \leftarrow (\tilde{\mathbf{W}}^{(t)} + \tilde{\mathbf{V}}^{(t)}) \mathbf{Z}^{(t)}$.

10: $t \leftarrow t + 1$.

11: **until** $|\sigma^{\text{geo}}(\mathbf{Z}^{(t)}) - \sigma^{\text{geo}}(\mathbf{Z}^{(t-1)})| < \varepsilon$ or $t \geq T$.

12: **return** $\mathbf{Z}^{(t)}$.

Initialization. We recommend initializing $\mathbf{Z}^{(0)}$ via classical MDS (eigendecomposition) applied to $\Delta^{(k)}$, which is the same initialization used by Isomap. This provides a warm start near the Isomap solution, from which the geodesic SMACOF iterations refine the embedding toward self-consistency.

Multiple restarts. Since the geodesic stress landscape is non-convex, multiple random restarts are advisable. In practice, we observe that the Isomap initialization converges to a good local minimum for most datasets, and random perturbations of the Isomap solution provide sufficient exploration.

Selection of k . The parameter k controls the tradeoff between tree-metric fidelity (small k) and ambient-metric fidelity (large k). We recommend selecting k by a cross-validation procedure: hold out a random fraction of edges from G_k , compute $\delta_{ij}^{(k)}$ on the remaining

graph for held-out pairs, and evaluate how well the embedding reproduces these held-out geodesics. The optimal k minimizes held-out geodesic stress. An alternative is to use the Gromov-hyperbolicity δ_0 of the input metric as a guide: for nearly tree-like metrics (small δ_0), small k is appropriate; for nearly Euclidean metrics (large δ_0), large k is preferable (Gromov, 1987).

3.8 A MISF-based Multiscale Modification of GMDS

The experiments of section 3.4 show that a global classical-MDS initialization can place GMDS into an unfavorable basin, especially on highly regular meshes. A natural alternative is to borrow the *maximal-independent-set filtration* (MISF) strategy introduced for large force-directed layouts by Gajer et al. (2004); Gajer and Kobourov (2002) and replace the force-directed objective at each scale by the geodesic objective developed above. The resulting method is still a fixed-geodesic GMDS method: the fidelity term remains eq. (11), but the initialization and the subsequent optimization are carried out in a coarse-to-fine manner.

Filtration. Given a connected weighted graph $G = (V, E, w)$, let

$$V = V_0 \supset V_1 \supset \cdots \supset V_L$$

be a maximal-independent-set filtration in the sense of Gajer et al. (2004); Gajer and Kobourov (2002): each $V_{\ell+1}$ is a maximal subset of V_ℓ whose vertices are separated in the graph by a prescribed minimum graph distance, so that the levels become progressively coarser. Intuitively, V_L is a sparse sample of the original graph, while $V_0 = V$ is the full vertex set. The filtration is shown on the underlying 2D mesh geometry in fig. 7. For the regular 12×12 paraboloid the extracted level sizes are $24 \rightarrow 50 \rightarrow 144$; for the regular 15×15 paraboloid they are $24 \rightarrow 28 \rightarrow 84 \rightarrow 225$; and for the irregular rectangle paraboloid they are $24 \rightarrow 32 \rightarrow 81 \rightarrow 225$.

Top level. Instead of initializing the whole graph by classical MDS, we first solve a *pure* GMDS problem only on the coarsest set V_L . Concretely, we form the complete graph on V_L with edge weights equal to the graph-geodesic distances $d_G(i, j)$ measured in the original graph G , choose a fixed geodesic family Γ_L on this induced metric graph, and minimize the raw geodesic stress on V_L . This coarse solve is the *initial embedding* of the MISF pipeline. The middle panel in the top row of figs. 8 to 10 shows this initial coarse embedding. It is not a random seed: it is already the result of a restartable pure-GMDS solve on V_L . The right panel in the same row shows that this coarse embedding is later revisited by the sparse refinement stage.

Geodesic insertion of finer levels. Once $V_{\ell+1}$ has been embedded, each new vertex $v \in V_\ell \setminus V_{\ell+1}$ is placed with respect to a small anchor set $A_\ell(v) \subseteq V_{\ell+1}$ drawn from the already embedded coarser level. The initial location of v is obtained by minimizing

$$\Phi_\ell(\mathbf{z}_v) = \sum_{a \in A_\ell(v)} \omega_{va} (\|\mathbf{z}_v - \mathbf{z}_a\| - d_G(v, a))^2,$$

where the anchor weights ω_{va} favor well-spread, geodesically close anchors. This is a local nonlinear least-squares problem and plays the role of geodesic trilateration. In the current

implementation each inserted vertex is optimized independently by a small Armijo line-search solver. The middle panels in the lower rows of figs. 8 to 10 show these insertion states before any active-set refinement.

Sparse active-set refinement. After all vertices of level V_ℓ have been inserted, the active set is refined by a sparse GMDS solve. Let $P_\ell \subseteq V_\ell \times V_\ell$ be a sparse pair set consisting of short-range geodesic neighbors together with a small number of landmark pairs. The level objective has the form

$$F_\ell(\mathbf{Z}) = \sum_{(i,j) \in P_\ell} (d_\Gamma^{\text{emb}}(i,j;\mathbf{Z}) - d_G(i,j))^2 + \lambda_\ell \sum_{u \notin V_\ell} \alpha_u \|\mathbf{z}_u - \bar{\mathbf{z}}_u\|^2,$$

where $\bar{\mathbf{z}}_u$ denotes the current scaffold position of an inactive vertex. The anchor term pins the already constructed coarser scaffold while allowing the active vertices to adjust. In other words, the method uses the same fixed-geodesic discrepancy as GMDS, but only on a sparse set of active pairs, and with the non-active scaffold softly constrained. The right panels of figs. 8 to 10 show the corresponding refinement states. After the levelwise cycle reaches V_0 , an optional short full-graph pure-GMDS polish is applied.

Implementation in grip. The current prototype reuses the GRIP infrastructure already present in the `grip` package. The MISF itself is extracted by `grip.build.misf()` in `R/grip.misf.R`, which delegates to the compiled GRIP filtration code in `src/DrawGraph.cpp` and `src/grip_rcpp.cpp`. The multiscale GMDS preparation and orchestration are implemented in `R/grip_geodesic.misf.R` through `grip.prepare.misf.geodesic.mds()` and `grip.optimize.misf.geodesic.mds()`. The per-vertex insertion kernel is implemented in `src/geodesic.misf_rcpp.cpp`, while both the coarse top-level solve and the sparse refinement stages reuse the compiled GMDS kernel in `src/geodesic.mds_rcpp.cpp`. Thus the MISF modification changes the optimization schedule and the initialization strategy, but not the underlying fixed-path geodesic discrepancy.

What the multiscale figures show. Figure 8 shows the regular 12×12 paraboloid. The coarsest initial embedding on V_2 is already far from a recognizable paraboloid sample ($\rho = 0.8588$), and the subsequent refinement improves it only to $\rho = 0.7109$. At the full level V_0 , the insertion state is highly tangled and refinement reduces the active-set error from $\rho = 0.6772$ to $\rho = 0.5763$, but the result is still not a smooth paraboloid. Figure 9 shows the same phenomenon on the regular 15×15 mesh: every level improves under refinement, yet the coarse sample and the full-level embedding remain visibly far from the target surface.

By contrast, fig. 10 shows that the irregular rectangle paraboloid is substantially easier for the same multiscale pipeline. Its coarsest initial embedding starts much closer to the target ($\rho = 0.5331$), and each subsequent level remains more consistent with the reference surface. The final level improves from $\rho = 0.3617$ after insertion to $\rho = 0.3343$ after sparse refinement, followed by a final full polish at $\rho = 0.3205$. This mirrors the empirical observation from the previous section: the difficulty is not merely the presence of curvature, but the interaction between the optimization landscape and the high symmetry of the regular rectangular mesh.

Table 1: Stagewise active-set Procrustes error ρ for the MISF-GMDS pipeline on the three paraboloid families shown in figs. 8 to 10. The “initial” column is the coarse top-level GMDS solve for the first active level and the anchor-based insertion state for the remaining levels; the “refined” column is the corresponding sparse GMDS refinement.

case	level	$ V_\ell $	initial ρ	refined ρ
regular 12×12	V_2	24	0.8588	0.7109
regular 12×12	V_1	50	0.8226	0.6731
regular 12×12	V_0	144	0.6772	0.5763
regular 12×12	final	144	—	0.5792
regular 15×15	V_3	24	0.9181	0.7114
regular 15×15	V_2	28	0.7335	0.6137
regular 15×15	V_1	84	0.6077	0.4897
regular 15×15	V_0	225	0.5709	0.4751
regular 15×15	final	225	—	0.4754
irregular 15×15	V_3	24	0.5331	0.4684
irregular 15×15	V_2	32	0.4970	0.4221
irregular 15×15	V_1	81	0.3969	0.3555
irregular 15×15	V_0	225	0.3617	0.3343
irregular 15×15	final	225	—	0.3205

3.9 Relationship to Metric Geometry

The construction of Geodesic MDS is grounded in several concepts from metric geometry that illuminate its properties.

Minimum realization. The k -NN graph G_k , after removing metrically redundant edges (pairs (i, j) for which $\delta_{ij}^{(k)}$ equals the shortest path through other vertices), yields the *minimum realization* of the input graph metric (Dress, 1984; Imrich et al., 1984). Geodesic MDS with this minimum realization as the underlying graph uses the most parsimonious geodesic structure consistent with the data.

Tight span. The *tight span* $T(\Delta)$ (or injective hull) (Dress, 1984; Isbell, 1964) is the minimal injective metric space containing $(X, \delta^{(k)})$. It is a polyhedral complex whose 1-skeleton contains the minimum realization. Geodesic MDS seeks an embedding of the 0-skeleton (the n data points) into $\mathbb{R}^d$ that preserves the 1-skeleton path metric, which is a projection of the tight span onto the original vertex set.

Metric spanners. As noted in section 2.6, G_k is a $(1 + \varepsilon)$ -spanner of the input Euclidean metric. Geodesic MDS with a spanner graph thus approximates the full Euclidean MDS problem with a multiplicative error of $(1 + \varepsilon)$ on input dissimilarities, while using a sparse graph structure that enforces geodesic consistency (Althöfer et al., 1993).

Gromov-Hausdorff approximation. The graph metric $(\{1, \dots, n\}, d_{G_k})$ converges in the Gromov-Hausdorff sense to the underlying geodesic metric space as $n \rightarrow \infty$ and k grows appropriately (Gromov, 1999; Latschev, 2001; Bernstein et al., 2000). Geodesic MDS can

therefore be interpreted as finding an optimal isometric embedding of a Gromov-Hausdorff approximation of the data geometry, rather than an optimal embedding of the ambient Euclidean point cloud.

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References

- Ingo Althöfer, Gautam Das, David Dobkin, Deborah Joseph, and José Soares. On sparse spanners of weighted graphs. *Discrete & Computational Geometry*, 9(1):81–100, 1993. doi: 10.1007/BF02189308.
- Mikhail Belkin and Partha Niyogi. Laplacian eigenmaps for dimensionality reduction and data representation. *Neural Computation*, 15(6):1373–1396, 2003. doi: 10.1162/089976603321780317.
- Mira Bernstein, Vin de Silva, John C. Langford, and Joshua B. Tenenbaum. Graph approximations to geodesics on embedded manifolds. *Technical Report, Department of Psychology, Stanford University*, 2000.
- Ingwer Borg and Patrick J. F. Groenen. *Modern Multidimensional Scaling: Theory and Applications*. Springer, New York, 2nd edition, 2005. doi: 10.1007/0-387-28981-8.
- Jean Bourgain. On Lipschitz embedding of finite metric spaces in Hilbert space. *Israel Journal of Mathematics*, 52(1–2):46–52, 1985. doi: 10.1007/BF02776078.
- Ronald R. Coifman and Stéphane Lafon. Diffusion maps. *Applied and Computational Harmonic Analysis*, 21(1):5–30, 2006. doi: 10.1016/j.acha.2006.04.006.
- Trevor F. Cox and Michael A. A. Cox. *Multidimensional Scaling*. Chapman and Hall/CRC, Boca Raton, FL, 2nd edition, 2000.
- Jan De Leeuw. Applications of convex analysis to multidimensional scaling. In J. R. Barra, F. Brodeau, G. Romier, and B. Van Cutsem, editors, *Recent Developments in Statistics*, pages 133–145. North-Holland, Amsterdam, 1977.
- Jan De Leeuw and Patrick Mair. Multidimensional scaling using majorization: SMACOF in R. *Journal of Statistical Software*, 31(3):1–30, 2009. doi: 10.18637/jss.v031.i03.
- Edsger W. Dijkstra. A note on two problems in connexion with graphs. *Numerische Mathematik*, 1(1):269–271, 1959. doi: 10.1007/BF01386390.
- David L. Donoho and Carrie Grimes. Hessian eigenmaps: Locally linear embedding techniques for high-dimensional data. *Proceedings of the National Academy of Sciences*, 100(10):5591–5596, 2003. doi: 10.1073/pnas.1031596100.

- Andreas W. M. Dress. Trees, tight extensions of metric spaces, and the cohomological dimension of certain groups: a note on combinatorial properties of metric spaces. *Advances in Mathematics*, 53(3):321–402, 1984. doi: 10.1016/0001-8708(84)90029-X.
- Carl Eckart and Gale Young. The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3):211–218, 1936. doi: 10.1007/BF02288367.
- Robert W. Floyd. Algorithm 97: Shortest path. *Communications of the ACM*, 5(6):345, 1962. doi: 10.1145/367766.368168.
- Michael L. Fredman and Robert E. Tarjan. Fibonacci heaps and their uses in improved network optimization algorithms. *Journal of the ACM*, 34(3):596–615, 1987. doi: 10.1145/28869.28874.
- Thomas M. J. Fruchterman and Edward M. Reingold. Graph drawing by force-directed placement. *Software: Practice and Experience*, 21(11):1129–1164, 1991. doi: 10.1002/spe.4380211102.
- Pawel Gajer and Stephen G. Kobourov. GRIP: Graph drawing with intelligent placement. *Journal of Graph Algorithms and Applications*, 6(3):203–224, 2002. doi: 10.7155/jgaa.00052.
- Pawel Gajer, Michael T. Goodrich, and Stephen G. Kobourov. A multi-dimensional approach to force-directed layouts of large graphs. *Computational Geometry*, 29(1):3–18, 2004. doi: 10.1016/j.comgeo.2004.03.014.
- John C. Gower. Some distance properties of latent root and vector methods used in multivariate analysis. *Biometrika*, 53(3–4):325–338, 1966. doi: 10.1093/biomet/53.3-4.325.
- Mikhail Gromov. Hyperbolic groups. In S. M. Gersten, editor, *Essays in Group Theory*, pages 75–263. Springer, New York, 1987. doi: 10.1007/978-1-4613-9586-7_3.
- Mikhail Gromov. *Metric Structures for Riemannian and Non-Riemannian Spaces*. Birkhäuser, Boston, MA, 1999. Based on the 1981 French original.
- Sariel Har-Peled and Manor Mendel. Fast construction of nets in low-dimensional metrics and their applications. *SIAM Journal on Computing*, 35(5):1148–1184, 2006. doi: 10.1137/S0097539704446684.
- Wilfried Imrich, J. M. S. Simoes-Pereira, and Christina M. Zamfirescu. On optimal embeddings of metrics in graphs. *Journal of Combinatorial Theory, Series B*, 36(1):1–15, 1984. doi: 10.1016/0095-8956(84)90009-0.
- J. R. Isbell. Six theorems about injective metric spaces. *Commentarii Mathematici Helvetici*, 39:65–76, 1964. doi: 10.1007/BF02566944.
- Tomihisa Kamada and Satoru Kawai. An algorithm for drawing general undirected graphs. *Information Processing Letters*, 31(1):7–15, 1989. doi: 10.1016/0020-0190(89)90102-6.
- Joseph B. Kruskal. Multidimensional scaling by optimizing goodness of fit to a nonmetric hypothesis. *Psychometrika*, 29(1):1–27, 1964a. doi: 10.1007/BF02289565.

- Joseph B. Kruskal. Nonmetric multidimensional scaling: a numerical method. *Psychometrika*, 29(2):115–129, 1964b. doi: 10.1007/BF02289568.
- Janko Latschev. Riemannian manifolds are not always well-approximated by their Vietoris–Rips complexes. *Geometriae Dedicata*, 88(1):295–307, 2001. doi: 10.1023/A:1013168001648.
- Nathan Linial, Eran London, and Yuri Rabinovich. The geometry of graphs and some of its algorithmic applications. *Combinatorica*, 15(2):215–245, 1995. doi: 10.1007/BF01200757.
- Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018. URL <https://arxiv.org/abs/1802.03426>.
- Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011. URL <https://jmlr.org/papers/v12/pedregosa11a.html>.
- Sam T. Roweis and Lawrence K. Saul. Nonlinear dimensionality reduction by locally linear embedding. *Science*, 290(5500):2323–2326, 2000. doi: 10.1126/science.290.5500.2323.
- John W. Sammon. A nonlinear mapping for data structure analysis. *IEEE Transactions on Computers*, 18(5):401–409, 1969. doi: 10.1109/T-C.1969.222678.
- Isaac J. Schoenberg. Remarks to Maurice Fréchet’s article “Sur la définition axiomatique d’une classe d’espace distanciés vectoriellement applicable sur l’espace de Hilbert”. *Annals of Mathematics*, 36(3):724–732, 1935. doi: 10.2307/1968654.
- Roger N. Shepard. The analysis of proximities: Multidimensional scaling with an unknown distance function. I. *Psychometrika*, 27(2):125–140, 1962. doi: 10.1007/BF02289630.
- Joshua B. Tenenbaum, Vin de Silva, and John C. Langford. A global geometric framework for nonlinear dimensionality reduction. *Science*, 290(5500):2319–2323, 2000. doi: 10.1126/science.290.5500.2319.
- Warren S. Torgerson. Multidimensional scaling: I. Theory and method. *Psychometrika*, 17(4):401–419, 1952. doi: 10.1007/BF02288916.
- Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, 2008. URL <https://jmlr.org/papers/v9/vandermaten08a.html>.
- Gale Young and Alston S. Householder. Discussion of a set of points in terms of their mutual distances. *Psychometrika*, 3(1):19–22, 1938. doi: 10.1007/BF02287916.
- Hongyuan Zha and Zhenyue Zhang. Continuum Isomap for manifold learnings. *Computational Statistics & Data Analysis*, 52(1):184–200, 2003. doi: 10.1016/j.csda.2006.11.027.

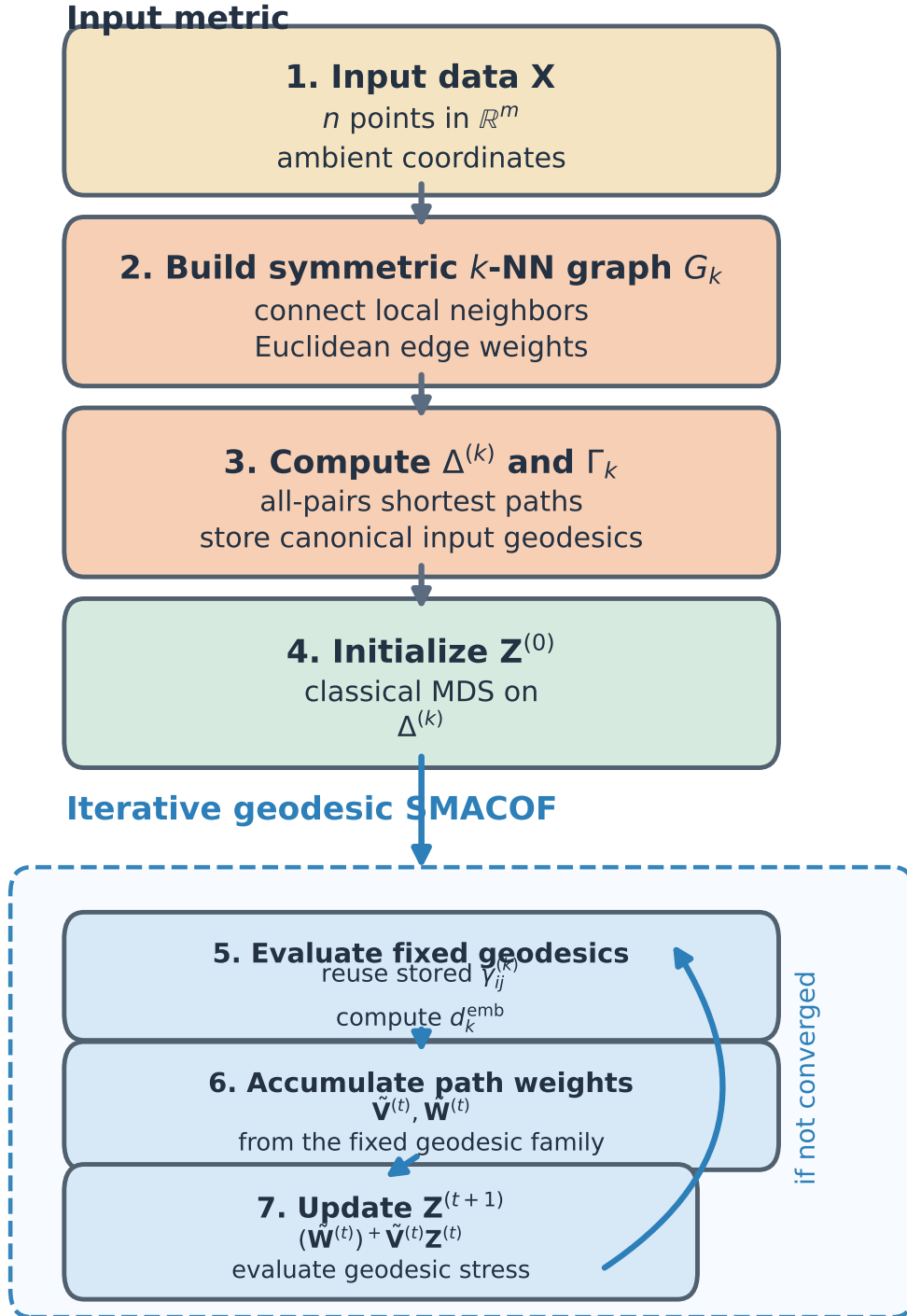


Figure 6: One iteration of Geodesic MDS. A canonical family of shortest paths is chosen once from the input graph. At each iteration, their embedded lengths are evaluated, converted into path-edge weights, and used to form a path-augmented Guttman update. The process begins from a classical-MDS initialization on the input graph geodesics and repeats until convergence.

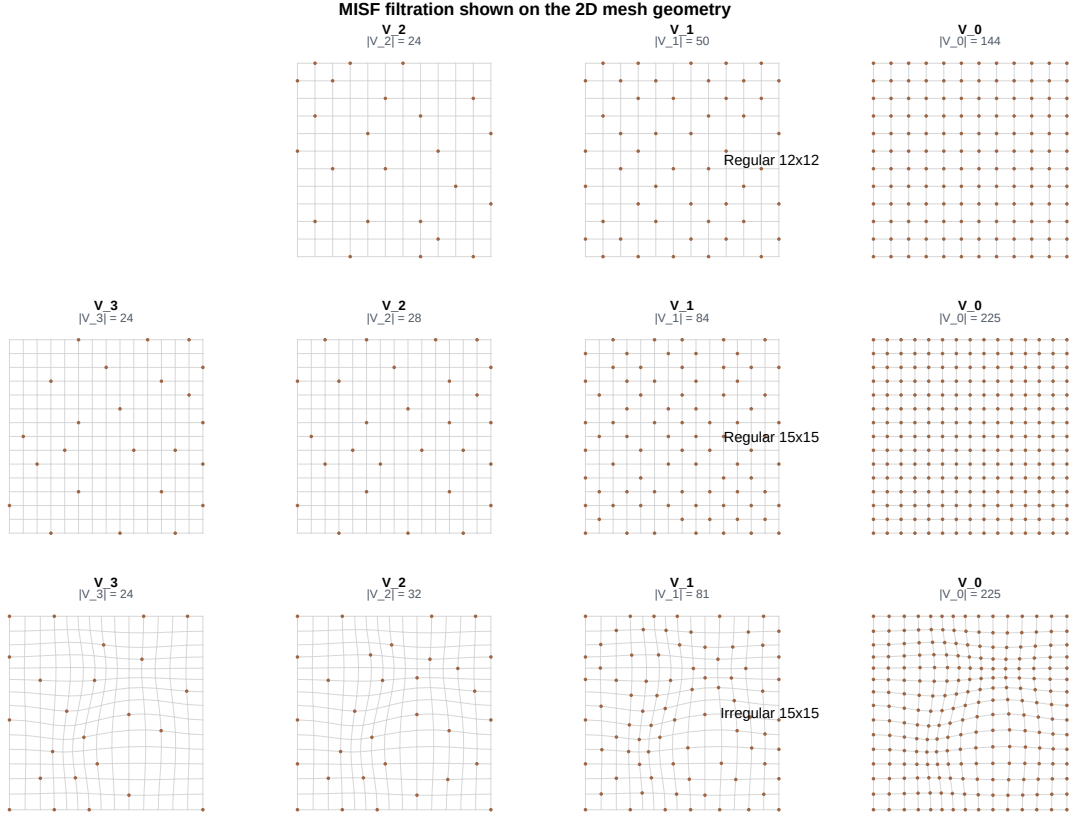


Figure 7: MISF levels displayed on the underlying 2D mesh geometry. Each row shows one graph family, with the active level V_ℓ highlighted on top of the full mesh graph. The irregular rectangle keeps rectangular topology but breaks the lattice symmetry in the geometry, yielding a more generic coarse sample than the perfectly regular meshes.

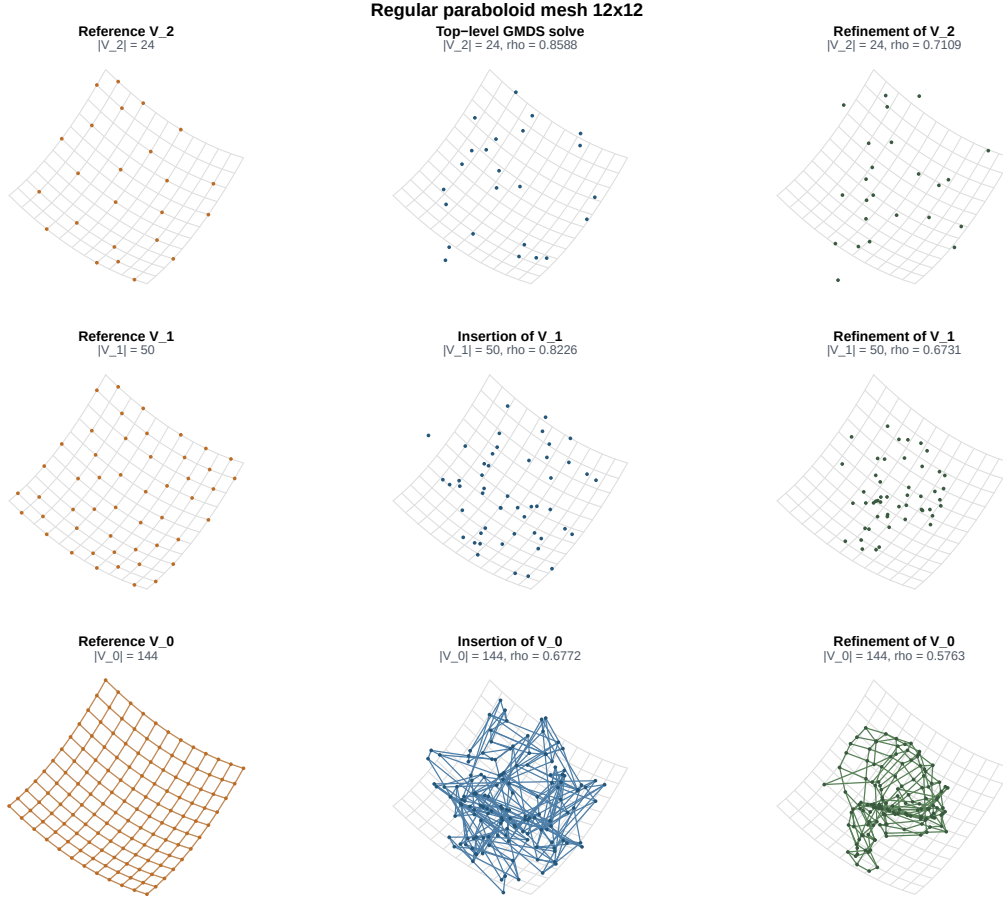


Figure 8: Regular 12×12 paraboloid under MISF-GMDS. Each row corresponds to one active set V_ℓ . Left: the reference active-set sample. Middle: the initial multiscale embedding for that level (the coarse pure-GMDS solve for V_2 , then anchor-based insertion for V_1 and V_0). Right: the sparse GMDS refinement of the same active set. The coarsest level is already not paraboloid-like, and although every level improves, the full embedding remains folded.

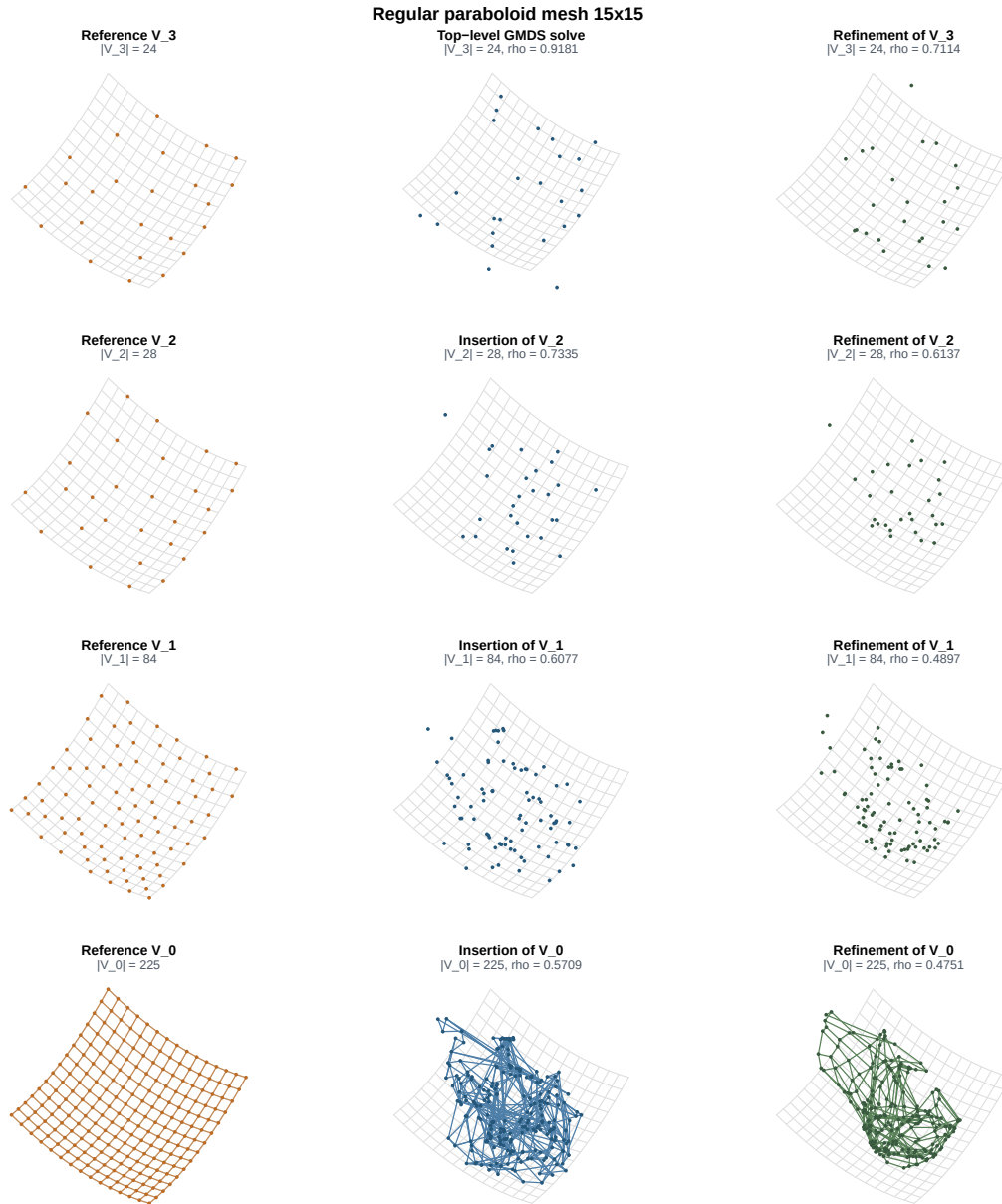


Figure 9: Regular 15×15 paraboloid under MISF-GMDS. The same qualitative pattern persists at the larger size: the coarsest-level solve does not recover a recognizable paraboloid sample, insertion creates a tangled full-level state, and sparse refinement improves but does not remove the pathology.

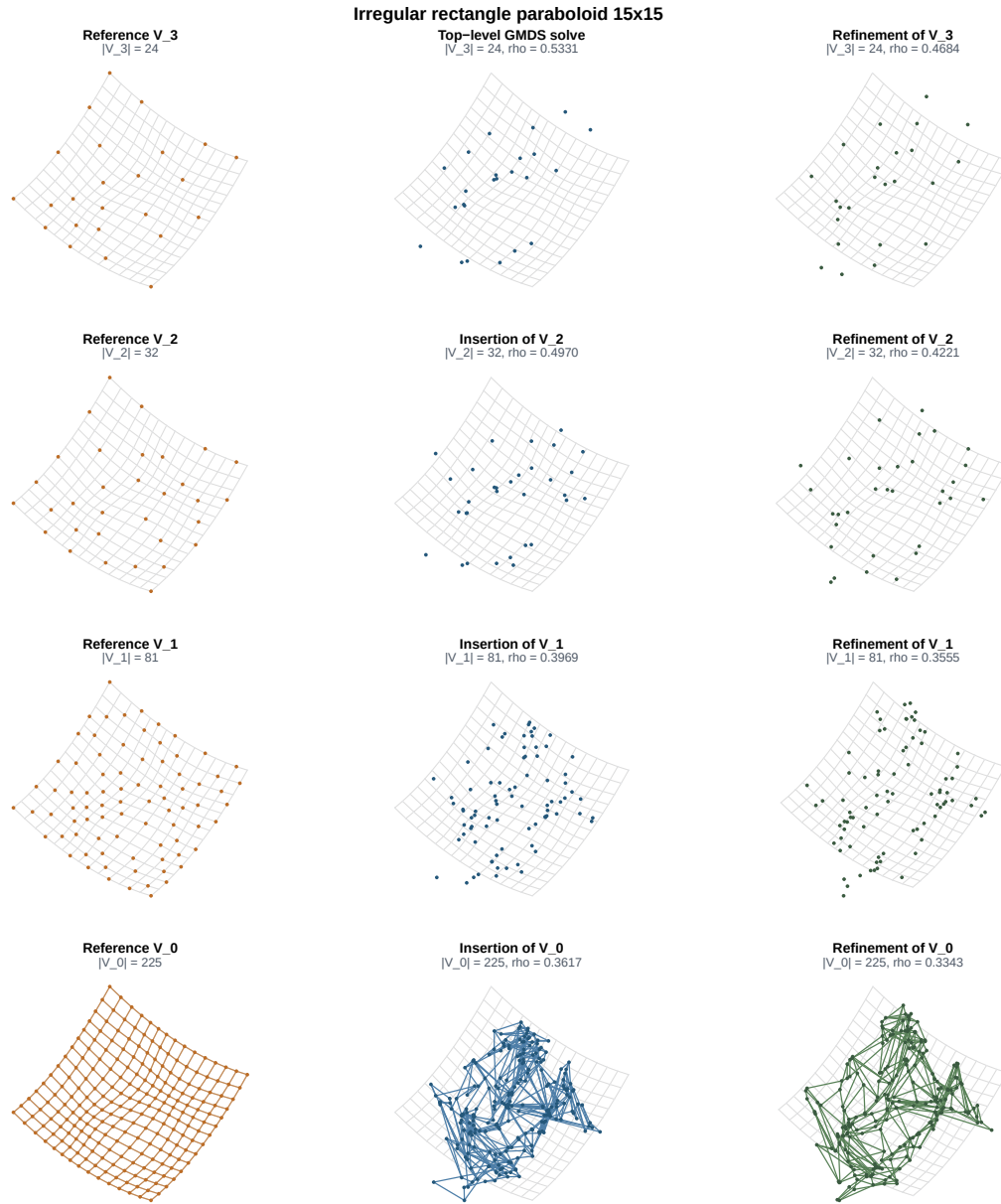


Figure 10: Irregular 15×15 rectangle lifted to a paraboloid. The multiscale pipeline is much better behaved on this less symmetric graph: the coarsest level already resembles a sparse paraboloid sample, and each finer refinement remains visibly closer to the reference geometry than in the regular-grid cases.